Spotiflow.

Accurate and efficient spot detection for fluorescence microscopy

With deep stereographic flow regression.
From noisy pixels to precise molecular coordinates.

Dominguez Mantes et al. · *Nature Methods* 22, 1495–1504

01 / DETECT

Multiscale heatmaps

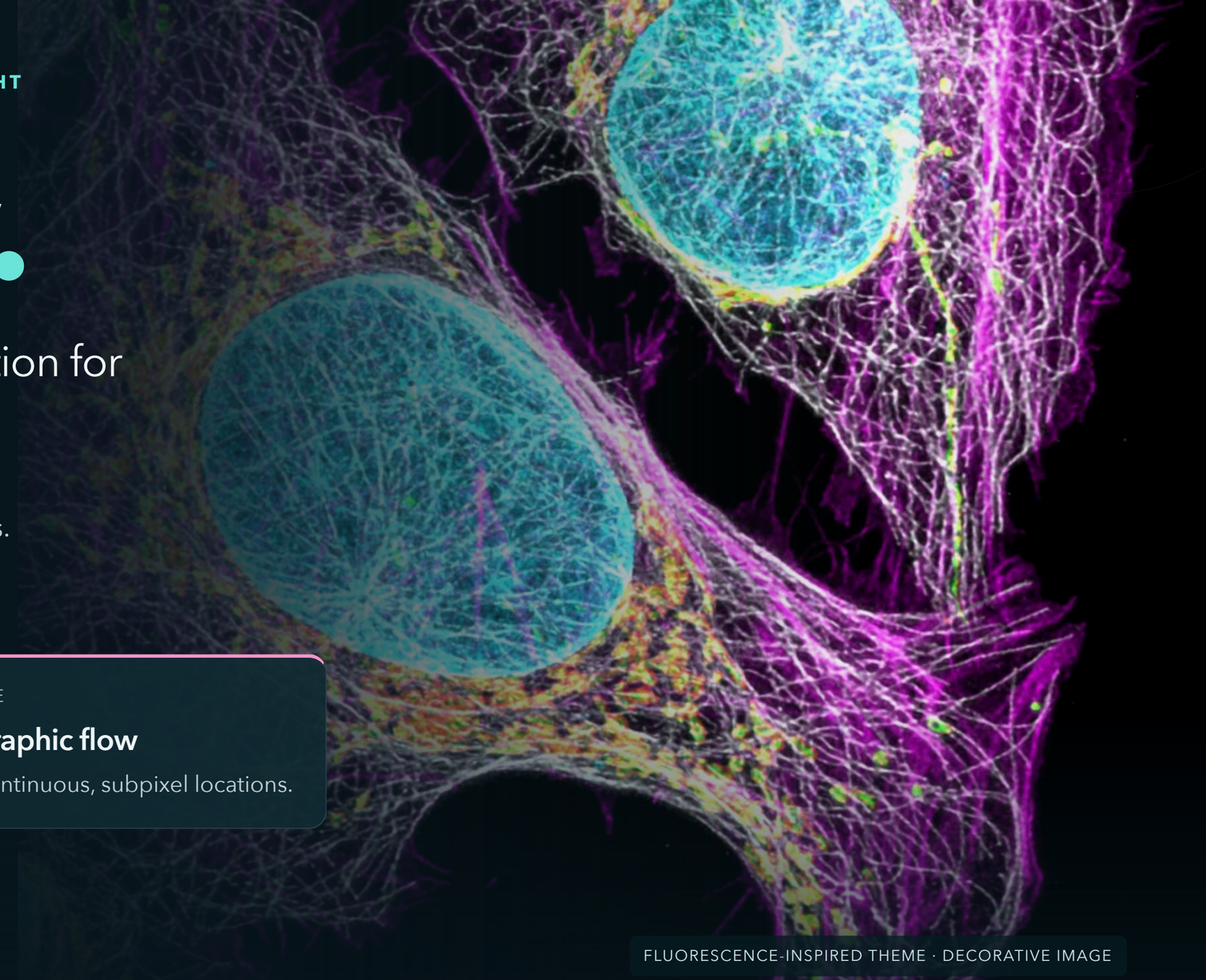
Identify candidate spots in 2D and 3D.

02 / REFINE

Stereographic flow

Recover continuous, subpixel locations.

Read the paper ↗ Code & documentation ↗ napari plugin ↗



FLUORESCENCE-INSPIRED THEME · DECORATIVE IMAGE

A missed spot can become a missed transcript.

Imaging-based spatial transcriptomics (iST) decodes RNA identity across sequential rounds and channels.

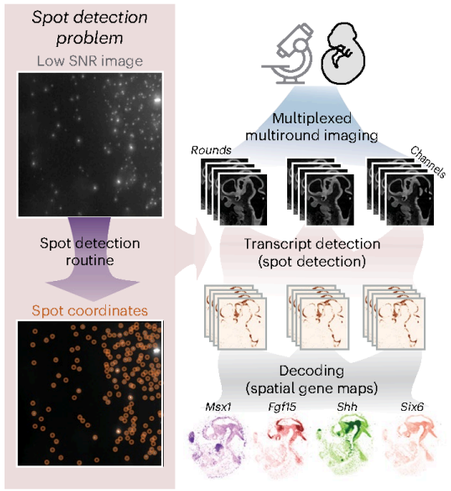


Fig. 1a · Detection is an early step in a multi-round decoding workflow.

Local error

A missed RNA spot or a spurious background detection.

→

Downstream effect

Reduced sensitivity or incorrect transcript assignment.

Millions

of spots in large, often gigabyte-sized images.

Real tissue is not a clean benchmark

- Autofluorescence & non-specific binding
- Uneven illumination and background
- Variable spot size and density

The goal is a reliable biological readout—not just a cleaner detection map.

Fig. 1a · Extended Data Fig. 6

Not just a new detector. A more complete workflow.

The paper positions Spotiflow at the intersection of robustness, precision, scale, and volumetric imaging.

METHOD FAMILY	ACKNOWLEDGED STRENGTHS	LIMITATIONS HIGHLIGHTED BY THE PAPER
Classical detectors LoG–Starfish · Big-FISH · RS-FISH	Effective on many clean or simulated images.	Sensitive to background artifacts, parameter choices, and dataset-specific thresholds.
Earlier deep learning SpotLearn · Detnet · deepBlink	Learn image features. **deepBlink supports subpixel localization.**	Usability and scaling gaps; most lack subpixel localization; no native 3D inference in the compared learning methods.
Spotiflow	Subpixel coordinates, scalable inference, and native 2D/3D support.	Requires suitable annotations; substantial domain changes may still require fine-tuning.

> The contribution is the **combination of capabilities**, not the claim that previous approaches never work.

One compact U-Net. Two complementary outputs.

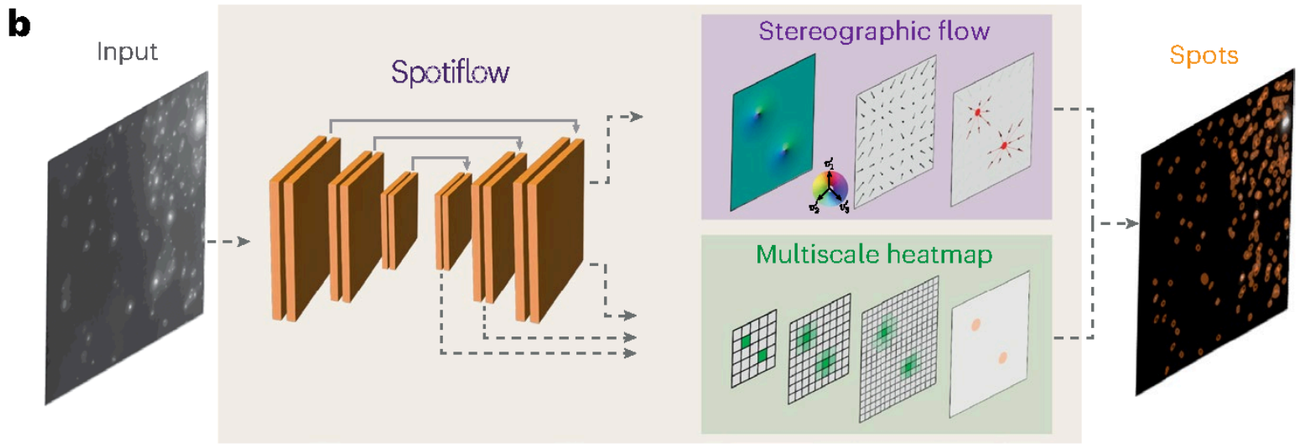


Fig. 1b · Shared features support detection and coordinate refinement.

$$\mathcal{L}(x) = (1 + \lambda \cdot \mathbb{1}_{\text{Spot}}(x))\,(\mathcal{L}_{\text{heat}}(x) + \mathcal{L}_{\text{flow}}(x))$$

Conceptual per-pixel weighting · $\lambda = 10$

The spot mask comes from the full-resolution target: $Y^{(0)}(x) \geq 0.01$.

SHARED BACKBONE

Compact 2D / 3D U-Net

Feature-map widths: **16, 32, 64**.
The reported d = 3 is convolutional-block depth, not automatically three downsampling stages.

Heatmap head

Spot probability at multiple resolutions.

Flow head

Three components in 2D.
Four components in 3D.

11× weight near annotated spots helps counter background imbalance.

Fig. 1b · Extended Data Fig. 1

Find the candidate.

A heatmap answers:
“Where is a spot likely to be?”

01

Use a **maximum**, not a sum, to form the target.

Multiscale Gaussian heatmaps

$$Y^{(0)}(x) = \max_{p_k} \exp \frac{-\|p_k - x\|^2}{2\sigma^2}$$

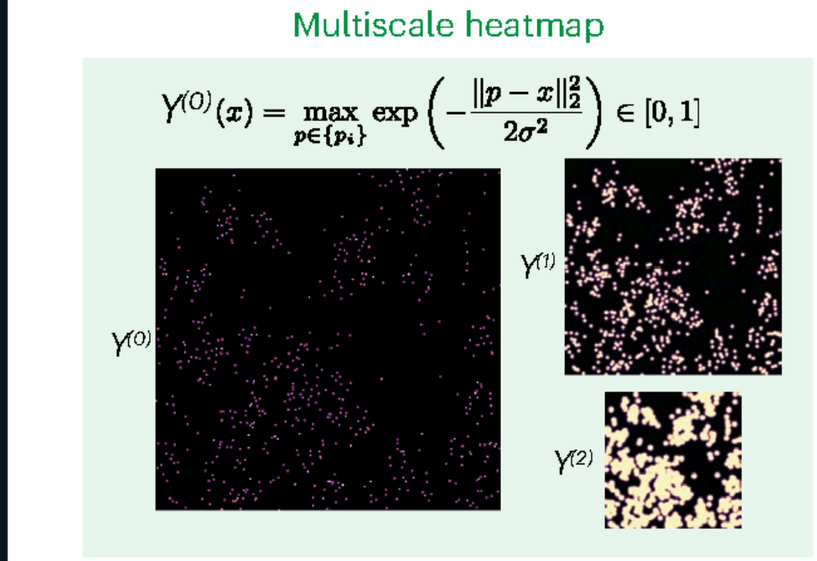


Fig. 1c · Supervision at multiple resolutions.

Preserve nearby peaks

A pixel-wise maximum avoids accumulating overlapping Gaussian responses into a broad plateau.

Build a target pyramid

Max-pooling, followed by Gaussian filtering, forms coarser targets.

Weight each resolution

Binary cross-entropy at level *l*, weighted by **1 / 2^l**.

Fig. 1c · Extended Data Fig. 1b

Turn unbounded offsets into bounded targets.

Map the offset to the nearest spot from $\mathbb{R}^2$ onto the unit sphere $S^2 \subset \mathbb{R}^3$.

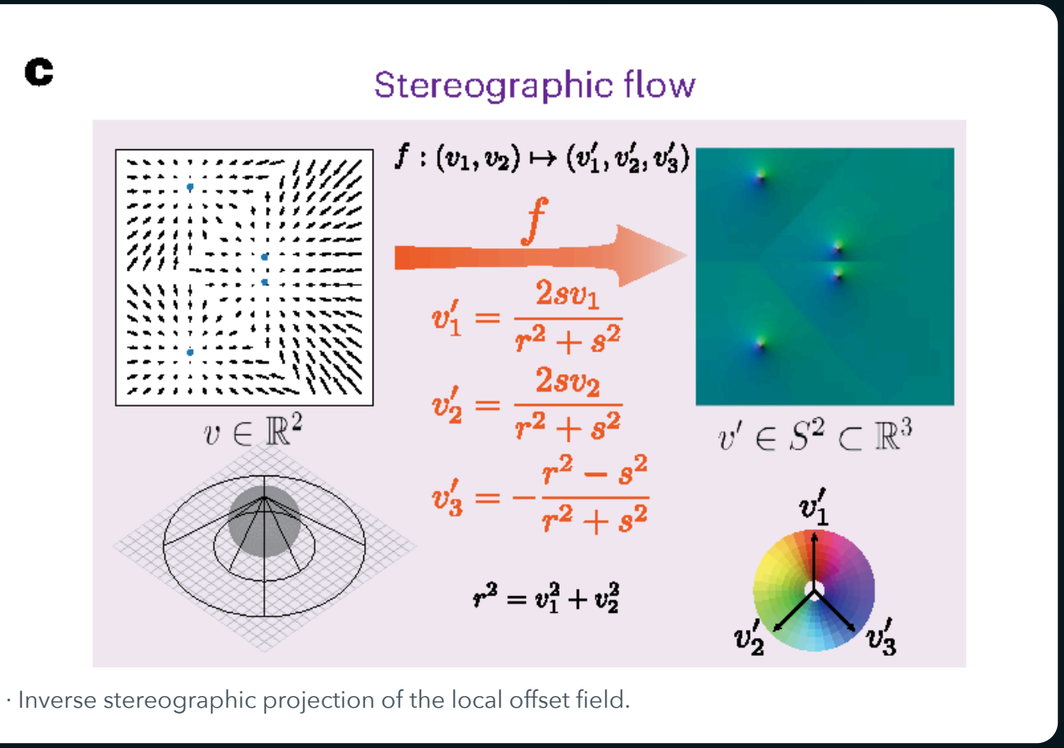


Fig. 1c · Inverse stereographic projection of the local offset field.

Zero offset

North pole: (0, 0, 1)

Offset → ∞

Approaches the south pole.

$$v_{ij} = p_{k^*} - (i, j)$$
$$k^* = \arg\min_k \|p_k - (i, j)\|_2$$

$$v'_x = \frac{2sv_x}{r^2 + s^2} \quad v'_y = \frac{2sv_y}{r^2 + s^2}$$
$$v'_z = -\frac{r^2 - s^2}{r^2 + s^2}$$

$r^2 = v_x^2 + v_y^2 \cdot s = 1$ in the paper

The benefit is **bounded magnitude**, making distant background offsets easier to handle.

Not globally continuous: nearest-spot assignments can still change across Voronoi boundaries.

Fig. 1c · Extended Data Fig. 2

Detect on the grid. Refine beyond the grid.

A continuous coordinate is recovered analytically from the flow at each heatmap candidate.

01 / CANDIDATES

Find local maxima

Use the highest-resolution probability heatmap.

→

02 / FILTER

Apply the cutoff

The model's threshold is selected on validation data.

→

03 / INVERT

Sample the flow

Recover a local offset from the predicted sphere vector.

→

04 / REFINE

Add the offset

Return subpixel coordinates instead of integer pixels.

ANALYTICAL INVERSE

$$\hat{v}_x = \frac{s\,\hat{v}'_x}{1+\hat{v}'_z} \qquad \hat{v}_y = \frac{s\,\hat{v}'_y}{1+\hat{v}'_z}$$

COORDINATE REFINEMENT

$$\hat{p}_k = (x_k, y_k) + \hat{v}_{x_k y_k}$$

Integer peak + learned fractional displacement

No per-image manual threshold tuning is required by the intended inference workflow.

The inverse is singular at the south pole. A good heatmap candidate should be near its spot, but neither a ≤0.5-pixel error nor a near-north-pole prediction is guaranteed.

The same geometry, one dimension higher.

3D offsets → four-component flow

$\mathbb{R}^3 \to S^3 \subset \mathbb{R}^4$ · a point on the unit 3-sphere.

$$(v'_x, v'_y, v'_z, v'_w) = \left(\frac{2sv_x}{r^2+s^2}, \frac{2sv_y}{r^2+s^2}, \frac{2sv_z}{r^2+s^2}, -\frac{r^2-s^2}{r^2+s^2}\right)$$

$r^2 = v_x^2 + v_y^2 + v_z^2$

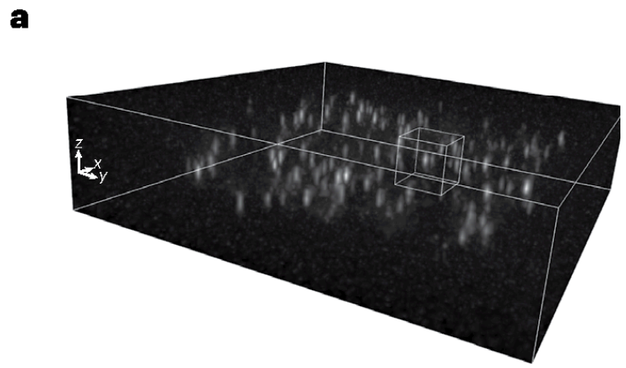


Fig. 4a · A native volumetric detection task.

GRIDDING / G = 2

Predict on a coarser grid

A strided 3D convolution reduces the output grid along each spatial dimension. Flow retains the fractional remainder.

RECOVER THE ORIGINAL COORDINATE

$p_H = (1, 1, 7), \quad \hat{v} = (0.5, 0, 0.5)$

$\hat{p} = g(p_H + \hat{v})$

$= 2(1.5, 1, 7.5) = (3, 2, 15)$

~10×

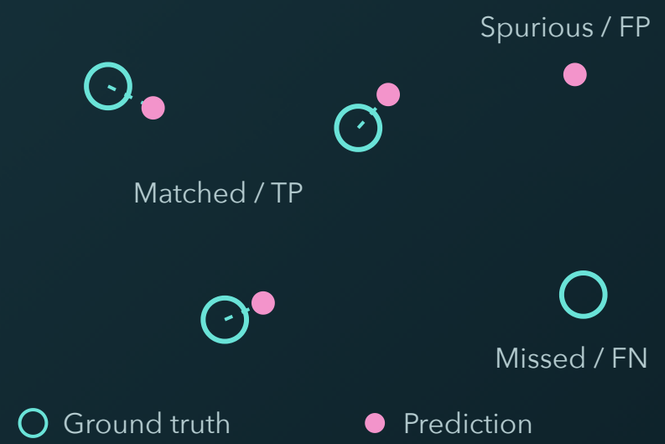
reported speed-up with g = 2.

Trade-off: nearby spots closer than the grid spacing become harder to distinguish.

Fig. 4 · Methods: gridding

Measure the right spots—and the right coordinates.

Unique spatial matching



Hungarian matching; a pair is a true positive when distance **d ≤ c**.

F1	TP / [TP + ½(FP + FN)] Balances missed and spurious detections.
AP	TP / (TP + FP + FN) The paper's Jaccard-like score, not conventional average precision.
F1 AuC	Area under F1 versus distance cutoff, from 1 to 5 pixels.
LA	Mean of **1 − d/c** over matched pairs. Mathematically consistent interpretation for d ≤ c.
PLQ	LA[c] × F1[c] Joint detection and localization quality.

Main cutoff: **c = 3 px**
Localization sweep: **1–5 px**

Apparent typo in published Eq. 12: max(d, c) would force LA = 0 for every matched pair. The consistent cutoff form uses min(d, c), equivalent to 1 − d/c for true positives.

The gap widens when the images become realistic.

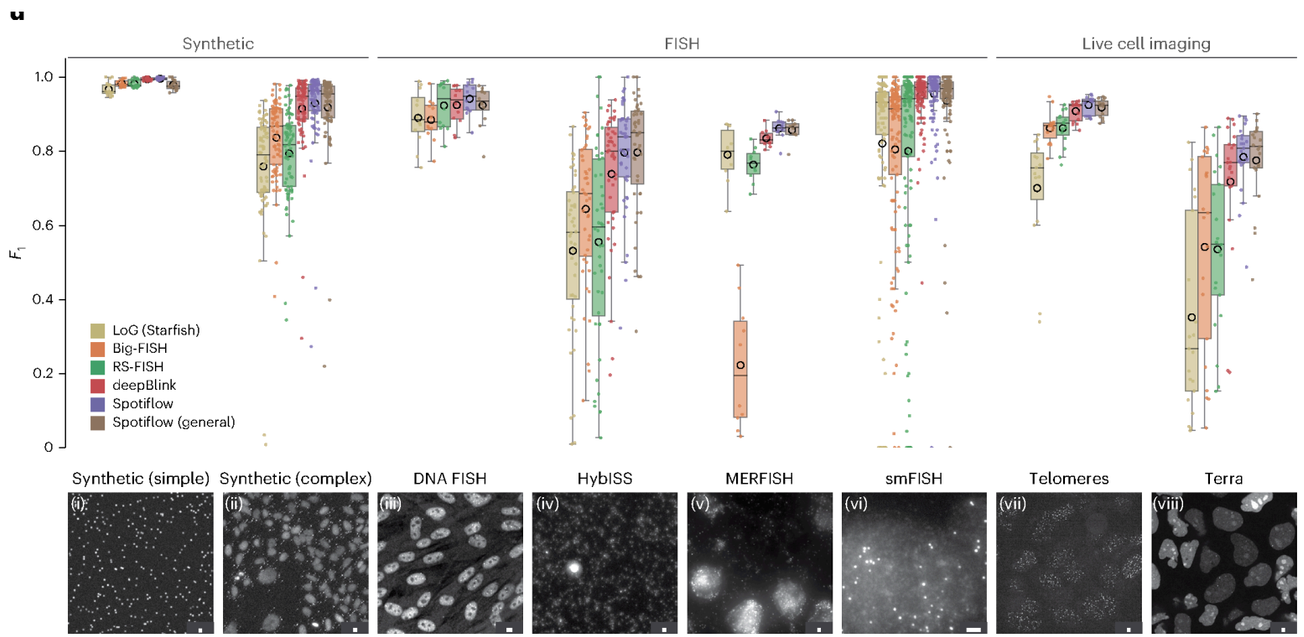


Fig. 1d · Synthetic, tissue, and live-cell benchmarks. Full method legend is included.

Clean data is easy for everyone

Synthetic-Simple F1 scores cluster around **0.967–0.995**.

TERRA / 4 TRAINING IMAGES

0.738 vs 0.174 F1

Fine-tuning versus training from scratch.

DNA FISH / 3 TRAINING IMAGES

0.77 → 0.92 F1

The general model is fine-tuned here—not a zero-shot result.

SYNTHETIC-COMPLEX

0.929

vs 0.915 deepBlink; 0.758–0.836 classical.

HYBISS

0.796

vs 0.531 LoG–Starfish.

MERFISH

0.861

vs 0.790 LoG–Starfish.

The general model was trained on the combined datasets except DNA FISH. F1 tolerates substantial annotation omissions; shifted labels can reduce localization quality.

Fig. 1d,f · Annotation robustness experiments

From spot coordinates to spatial gene expression.

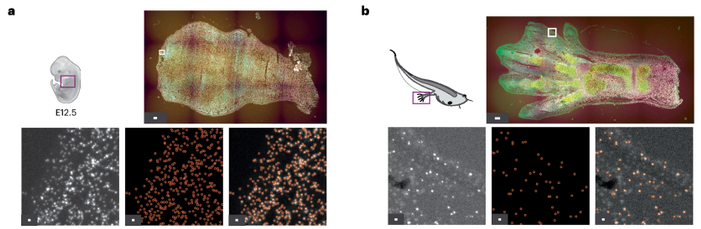


Fig. 2a,b · Transfer to other samples without retraining.

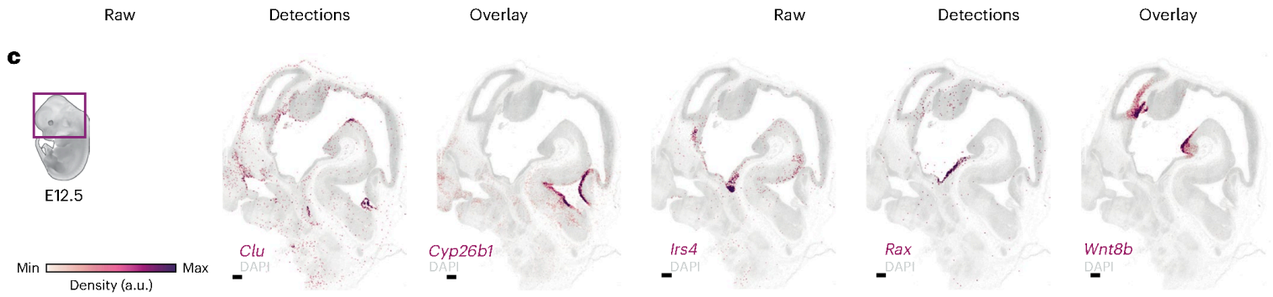


Fig. 2c · Selected gene patterns agree with in situ hybridization references.

199 **neurodevelopment-related genes**

End-to-end Starfish decoding of E12.5 and E13.5 mouse-brain HybISS data.

Transfer across samples

Embryonic mouse-brain training → developing mouse limb and frog tadpole limb images.

Less manual threshold sensitivity

Spotiflow uses a validation-selected cutoff; LoG–Starfish output varies with manually chosen detection thresholds.

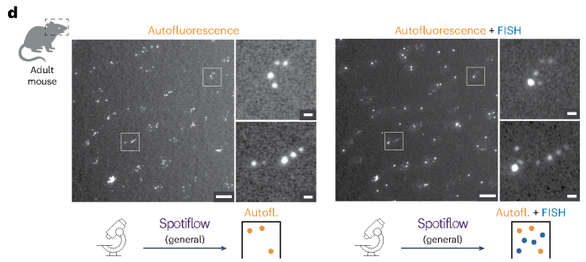
Benchmark gains are connected to a **downstream biological task**.

Fig. 2

Handle background. Enter volumes. Follow motion.

FIG. 3 / AUTOFLUORESCENCE

Learn what not to detect



Spot-like lipofuscin can resemble RNA.

An autofluorescence round identifies likely background spots for fine-tuning.

Fewer autofluorescent detections than intensity thresholding at **matched non-autofluorescent counts**.

FIG. 4 / NATIVE 3D

Volumetric context matters

SYNTHETIC-3D / F1

Spotiflow 3D	0.882
Big-FISH 3D	0.683
Spotiflow 2D	0.311
deepBlink 2D	0.102

2D methods use Z-blending; bars run from 0 to 1.

97% vs 91% spots identified at 1,600 spots/stack: raw-image Spotiflow versus deconvolution + intensity-based detection.

Fine-tuning on 9 annotated *P. dumerilii* subvolumes transfers to 159-GB mouse EASI-FISH data without further fine-tuning.

FIG. 3 / LIVE-CELL IMAGING

Support longer tracks

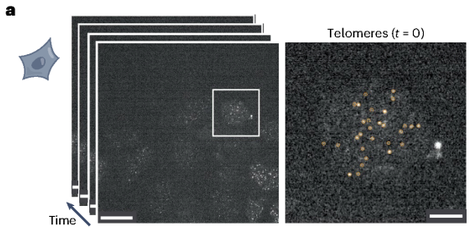


Fig. 3a · Spot detections support tracking.

In the tested live-cell applications, Spotiflow supports longer, less fragmented tracks.

The value extends from individual images to **dynamic biological measurements**.

Fig. 3: autofluorescence / tracking · Fig. 4: 3D

Built for images that do not fit a toy workflow.

32,000 × 32,000 PIXELS

~80 s

Spotiflow · RTX 4090 GPU

SAME IMAGE SIZE

~1,000 s

LoG–Starfish · 24-core CPU

Reported end-to-end timing on different hardware—not a hardware-controlled comparison of algorithmic speed.

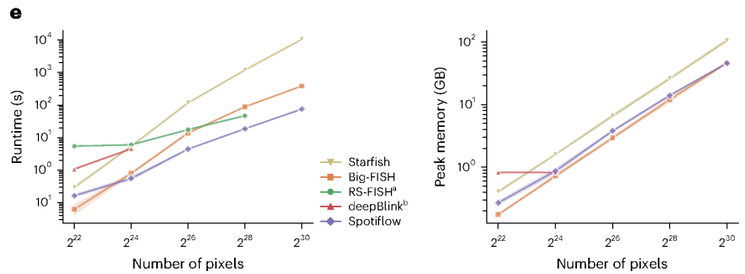


Fig. 1e · Runtime and memory scaling across implementations.

Compared implementations: deepBlink could not run above 4,096 pixels (GPU memory); RS-FISH was not shown above 32,768 pixels (Java size-related limits).

3D at scale

~1,600³ voxels: **<4 min** on one GPU.
159-GB EASI-FISH: **just under 1 h** on one GPU; **<10 min** on eight A100s.

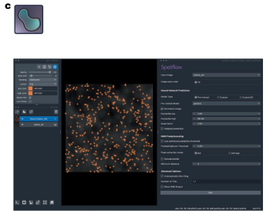


Fig. 5c · Interactive access through napari.

SOFTWARE ECOSYSTEM

Python API · CLI · napari

Plus Starfish integration for downstream transcript decoding.

Fig. 1e · Fig. 4f · Fig. 5

How does the paper make its case?

From technical contribution to scientific communication.



7

abstract sentences

~160 words

Build a problem funnel.

MOVE 1 / BROAD RELEVANCE

A recurring microscopy problem

Accurate spot detection matters across life-science imaging.

MOVE 2 / CONCRETE APPLICATION

Spatial transcriptomics raises the stakes

Very large numbers of transcripts, often in noisy images.

MOVE 3 / THE GAP

Current workflows still need tuning

Classical signal processing and dataset-specific settings remain common.

Why it works: establish relevance, narrow the context, then give readers a clear reason a new method is needed.

Resolve the problem in three deliberate moves.

An interpretation of the abstract's second half: solution → practical value → evidence and access.

MOVE 4

What?

Name the method and its idea

Spotiflow delivers subpixel localization through a two-part formulation.

> Heatmap detection
> + stereographic flow

MOVE 5

Why?

Translate method into utility

2D/3D support, generalization across imaging conditions, and time/memory efficiency.

> Capabilities that matter
> to actual workflows

MOVE 6

Proof?

Close with evidence and access

Quantitative benchmarks, downstream biological value, and usable software.

> Python library
> + napari plugin

The methods-paper promise: not only an algorithm that performs well, but a tool researchers can put into practice.

Define the task. Then explain why errors matter.

PARAGRAPH 1 / DEFINE EFFICIENTLY

Four questions, one coherent task

What?	RNA molecules.
Where?	In native tissue sections.
How?	Across sequential imaging rounds.
Why?	Gene-expression maps at subcellular resolution.

> Readers understand the measurement before they meet the model.

PARAGRAPH 2 / ESTABLISH CONSEQUENCES

Real tissue complicates detection

Autofluorescence, non-specific binding, and variable spot sizes and densities.

01 / A local spot-detection error

A false negative or false positive in one round or channel.

02 / A disrupted barcode

Transcript identity relies on a pattern across images.

03 / A biological consequence

Reduced sensitivity or incorrect transcript assignment.

A fair critique is a more persuasive critique.

Acknowledge what existing methods solve, then isolate the combination of requirements still unmet.

CLASSICAL SIGNAL PROCESSING

Strong on clean data

Effective on many clean or simulated images.

REMAINING NEED

Robustness to realistic tissue artifacts and changing contrast, with less dataset-specific tuning.

EARLIER DEEP LEARNING

Learned image representations

The paper explicitly credits **deepBlink's subpixel localization**.

REMAINING NEED

A usable workflow combining precision, large-image scaling, and native 3D support.

Concede the strength → identify the limit → state the unmet need.

• OUR RHETORICAL READING

Make every contribution answer a concrete need.

GAP ESTABLISHED EARLIER	HOW THE PAPER POSITIONS SPOTIFLOW
Dataset-specific threshold tuning	Validation-selected detection threshold
Integer-only / less precise localization	Stereographic flow for subpixel refinement
Very large images	Runtime and memory scaling experiments
2D-only learning workflows	Native 3D model and volumetric evaluation
Barriers to adoption	Python API · CLI · napari · Starfish integration

An interpretive mapping, not a strict one-to-one taxonomy claimed by the authors.

Technical takeaway

Multiscale heatmaps find the spots. Stereographic flow refines their coordinates.

• 2D + 3D · SUBPIXEL · SCALE

Reusable writing pattern: define the task → establish stakes → critique fairly → state the gap → connect claims to evidence.

Paper ↗ Code & documentation ↗ Data & annotations ↗